

Mensuration

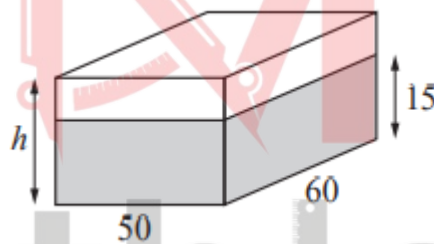
Worksheet 2a

Cube/Cuboid

1. S09/qp2/q7(a)

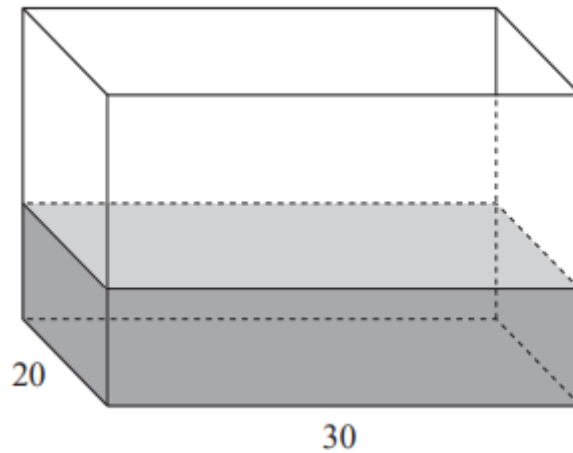
When a solid rectangular wooden block of oak floats, 60% of its height is under water.

- i. What fraction of its height is above water? [1]
- ii. A block of oak has length 60cm, breadth 50cm and height h centimetres. It floats with 15cm of its height under water.



- a. Find the value of h . [1]
- b. In the diagram, the shaded region represents part of the surface area of the block that is in contact with the water. Calculate the total surface area of the block that is in contact with the water. [2]

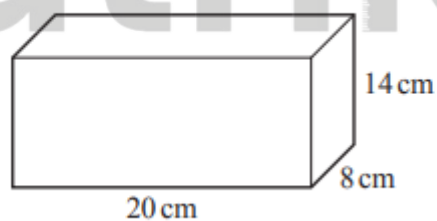
2. W10/qp21/q7a(i)(ii)(iii)



The diagram shows an open rectangular tank with base 20 cm by 30 cm.
The tank contains 9600 cm^3 of water.

- i. State the number of litres of water in the tank. [1]
- ii. Calculate the depth of the water. [2]
- iii. Calculate the **total** surface area of the tank that is in contact with the water. [2]

3. S15/qp13/q20(b) 0580

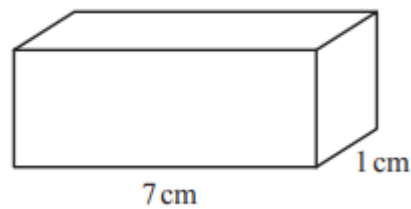


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- i. Calculate the surface area of this cuboid. [3]
- ii. Write your answer to part (b)(i) in square millimetres. [1]

4. S18/qp22/q14 0580



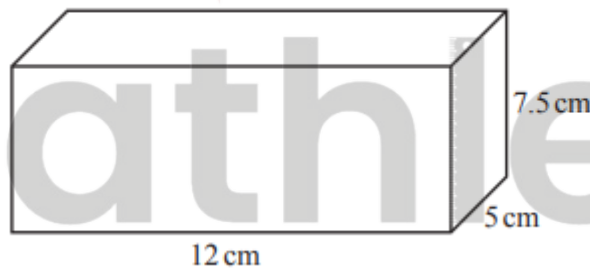
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The diagram shows a solid cuboid with base area 7cm^2 . The volume of this cuboid is 21cm^3 . Work out the total surface area. [3]

5. W18/qp21/q10 (0580)

A water tank in the shape of a cuboid has length 1.5 metres and width 1 metre. The water in the tank is 60 centimetres deep. Calculate the number of litres of water in the tank. [3]

6. M19/qp22/10 (0580)



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Calculate the total surface area of the cuboid. [3]

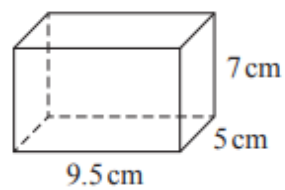
7. S19/qp41/q10(a) (0580)

The volume of each of the following solids is 1000cm^3 . Calculate the value of x for each solid: A cube with side length x cm. [1]

8. W19/qp23/q9 (0580)

A cuboid measures 5cm by 7cm by 9.5cm. Work out the surface area of this cuboid.

[3]



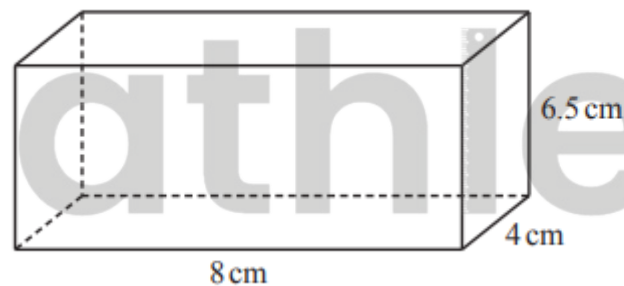
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9. M20/qp42/q3(c) (0580)

Water from Manjeet's shower flows at a rate of 12 litres per minute. The water from the shower flows into a tank that is a cuboid of length 90cm and width 75cm. Calculate the increase in the level of water in the tank when the shower is used for 7 minutes.

[3]

10. S20/qp23/q2 (0580)

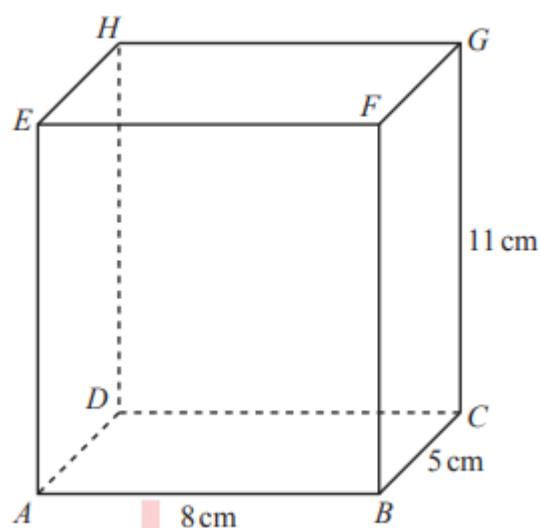


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The diagram shows a cuboid. Calculate the volume of the cuboid.

[1]

11. W20/qp43/q6 (0580)



ABCDEFGH is a cuboid. $AB = 8$ cm, $BC = 5$ cm and $CG = 11$ cm.

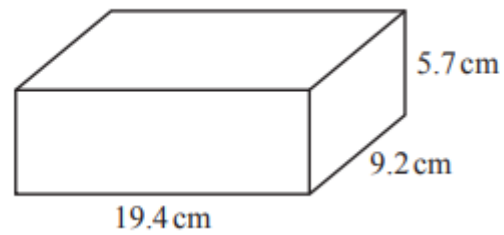
- a. Work out the volume of the cuboid. [2]
- b. Ivana has a pencil of length 13 cm. Does this pencil fit completely inside the cuboid? Show how you decide. [4]
- c.
 - i. Calculate angle CAB. [2]
 - ii. Calculate angle GAC. [2]

12. S21/qp43/q8(a) (0580)

A solid cuboid measures 20 cm by 12 cm by 5 cm.

- i. the volume of the cuboid. [1]
- ii.
 - a. Calculate the total surface area of the cuboid. [3]
 - b. The surface of the cuboid is painted. The cost of the paint used is \$1.52 . Find the cost to paint 1 cm^2 of the cuboid. Give your answer in cents. [1]

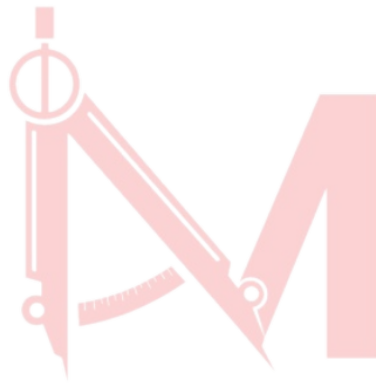
13. W21/qp41/q1(a) 0580



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The diagram shows a brick in the shape of a cuboid.

- i. Calculate the total surface area of the brick. [3]
- ii. The density of the brick is 1.9 g/cm^3 . Work out the mass of the brick.
Give your answer in kilograms. [Density = mass $\div$ volume] [3]



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Answer Key

1. S09/qp2/q7(a)

7	(a)	(i)	$\frac{2}{5}$ oe fraction	B1	1	Not 40%; 0.4
		(ii) (a)	$h = 25$	B1	1	
		(b)	$2(50 \times 15 + 60 \times 15) + 50 \times 60$ 6300 cm^2	M1		
				A1	2	sc1 for 3300 or for 9300 cm^2

2. W10/qp21/q7a(i)(ii)(iii)

7	(a)	(i)	9.6	1		
		(ii)	16 cm	2	M1 for $\frac{9600}{20 \times 30}$	
		(iii)	2 200 cm^2	2ft	B1 for areas 20×30 , their 16×20 and their 16×30 ft for $600 + 100 \times$ their (a)(ii)	
		(iv)	191	3	B1 for $\pi \times 0.8^2 \times 25$ soi M1 for their $(\pi \times 0.8^2 \times 25) \times t = 9600$	

3. S15/qp13/q20(b) 0580

(b)(i)	1104	3	M2 for $2 \times (20 \times 14 + 20 \times 8 + 8 \times 14)$ oe or M1 for $20 \times 14 + 20 \times 8 + 8 \times 14$
(ii)	their (b)(i) $\times 100$	1FT	

4. S18/qp22/q14 0580

14	62	3	M1 for [height =] $21 \div 7$ M1 for $2(1 \times \text{their}3 + \text{their}3 \times 7 + 1 \times 7)$ oe
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5. W18/qp21/q10 0580

10	900	3	M2 for $\frac{150 \times 100 \times 60}{1000}$ oe or M1 for $150 \times 100 \times 60$ or $1.5[\times 1] \times 0.6$ or B1 for figs 9
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6. M19/qp22/10 0580

10	375	3	M2 for $2(12 \times 5 + 12 \times 7.5 + 5 \times 7.5)$ oe or M1 for 12×5 or 12×7.5 or 5×7.5
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7. S19/qp41/q10(a) 0580

10(a)	10	1	
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8. W19/qp23/q9 0580

9	298	3	M2 for $[2 \times] (5 \times 7 + 5 \times 9.5 + 7 \times 9.5)$ oe or M1 for one correct area, 5×7 or 5×9.5 or 7×9.5
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9. M20/qp42/q3(c) 0580

3(c)	12.4 or 12.44...	3	M1 for $90 \times 75 \times h = 7 \times \text{figs } 12$ B1 for $1000 \text{ cm}^3 = 1 \text{ litre}$ soi
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10. S20/qp23/q2 0580

2	208	1
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11. W20/qp43/q6 0580

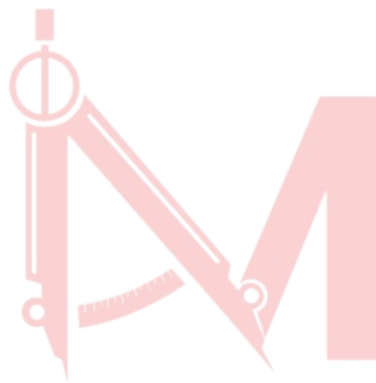
6(a)	440	2	M1 for $8 \times 5 \times 11$
6(b)	$\sqrt{8^2 + 5^2 + 11^2}$ oe or $8^2 + 5^2 + 11^2$ and 13^2 <u>ALTERNATIVE</u> $\sqrt{8^2 + 11^2}$ or $8^2 + 11^2$ and 13^2	M3	M2 for $8^2 + 5^2 + 11^2$ or $8^2 + 11^2$ oe or M1 for $8^2 + 5^2$ or $5^2 + 11^2$ oe
	Yes and 14.5 or 14.4 or 14.49... or Yes and 13.6[0...]	A1	Accept equivalent conclusion
6(c)(i)	32.0[...]	2	M1 for $\tan[.] = \frac{5}{8}$ oe
6(c)(ii)	49.4 or 49.38 to 49.39	2	M1 for $\sin[.] = \frac{11}{\text{their } AG}$ oe

12. S21/qp43/q8(a) 0580

8(a)(i)	1200	1	
8(a)(ii)(a)	800	3	M2 for $[2 \times] (20 \times 12 + 20 \times 5 + 12 \times 5)$ or M1 for 20×12 or 20×5 or 12×5
8(a)(ii)(b)	0.19	1	FT $152 \div \textit{their } 800$

13. W21/qp41/q1(a) 0580

1(a)(i)	683	3	M2 for $[2]((19.4 \times 9.2) + (5.7 \times 9.2) + (19.4 \times 5.7))$ oe or M1 for one of 19.4×9.2 or 5.7×9.2 or 19.4×5.7
1(a)(ii)	1.93[0] or 1.932 to 1.933	3	M2 for $19.4 \times 9.2 \times 5.7 \times 1.9$ or M1 for $19.4 \times 9.2 \times 5.7$



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